Runguo Li

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Research Interests

Machine learning systems for large models: efficient long-context LLM inference (sparse attention, KV-cache compression, quantization), systems for LLM agents, distributed training and RL post-training infrastructure, and the correctness of ML compilers and GPU kernels.

Education

University of Illinois Urbana-Champaign (UIUC) Aug 2026 – Present
M.S. in Information Management (expected Jul 2028) Urbana-Champaign, IL, USA

Shanghai University of Finance and Economics (SUFE) Sep 2022 – Jun 2026
B.S. in Business Analytics, GPA: 3.71/4.0 Shanghai, China

Publications and Preprints

(* equal contribution)

[1] **Runguo Li**. **Certified Sparse Attention via Perturbed Retrieval**. *arXiv preprint arXiv:2610.24991, 2026.* [arXiv] [PDF]
Sparse attention with per-head error certificates; 4.6–7.9× lower tail error than the best of 15 baseline families at 16K–128K tokens.

[2] **Runguo Li**. **Spend Bytes on Breadth: Precision–Count Trade-offs for Decode-Time KV Compression in Long Chain-of-Thought Reasoning**. *arXiv preprint arXiv:2610.05685, 2026.* [arXiv] [PDF]
Splits a fixed KV byte budget between token count and precision; beats eviction alone in 17 of 18 settings.

[3] **Runguo Li**. **How Much Does Random Rotation Help GPTQ? Sharp Worst-Case Rate Gaps in the High-Rate Regime**. *Under review, 2026.* [PDF]
Sharp worst-case rate gap between randomly rotated GPTQ and WaterSIC, answering an open question.

[4] **Runguo Li**. **AXR: Reusing Intermediate Results for LLM Agents in PostgreSQL**. *Under review, 2026.* [PDF]
PostgreSQL extension that reuses plan-subtree results across agent attempts; 6.5–51.0% less database time.

[5] Zhi Yang*, **Runguo Li***, et al. **FinVault: Benchmarking Financial Agent Safety in Execution-Grounded Environments**. *Under review, 2026; arXiv:2601.07853.* [arXiv] [Code]
Execution-grounded safety benchmark for financial agents (31 sandbox scenarios, 963 test cases).

[6] Yeheng Chen*, **Runguo Li***, et al. **VeriBRT: Plan-Guided, Evidence-Based Automated Bug Reproduction**. *Under review, 2026.* [PDF]
Plan-guided bug-reproducing test generation; 341 of 433 issues on SWT-Bench Verified.

[7] Yutao Hou*, **Runguo Li***, et al. **Reasoning-Visual Critical Token Fine-Tuning for Multimodal Reasoning**. *Under review, 2026.* [PDF]
Supervises only reasoning-relevant and visually sensitive chain-of-thought tokens.

Research Experience

University of Illinois Urbana-Champaign Aug 2026 – Present
Research Intern, advised by Prof. Minjia Zhang Urbana-Champaign, IL, USA

- Efficient long-context inference: certified sparse attention [1] and decode-time KV-cache compression for long chain-of-thought reasoning [2].

Shanghai Jiao Tong University Mar 2026 – Jul 2026
Research Intern, advised by Prof. Xiaodong Gu Shanghai, China

- LLMs for code: co-first author of **VeriBRT** [6] on automated bug reproduction; built the SWT-Bench Verified evaluation harness.

Tencent Youtu AI Lab Oct 2025 – May 2026
Research Intern Shanghai, China

- Co-first author of **RVCFT** [7] on multimodal chain-of-thought fine-tuning; adversarial data synthesis and multilingual content-safety moderation.

Open-Source Contributions

22 merged pull requests across 14 ML-systems projects (project names link to the merged PRs):

Inference and serving

- **vLLM-Omni**: block-sparse attention in the Qwen2.5-Omni Token2Wav DiT, 3.5–6.4× lower per-request latency on H200 (#7975, #7959).
- **vLLM**: fixed int4 per-token-head KV-cache quantization for non-power-of-two head sizes (#56198).
- **SGLang**: stopped DeepEP import-time failures from crashing servers that never use DeepEP (#40671).

Distributed training and post-training

- **Megatron-LM**: 165× faster MoE state-dict conversion and 3.4× shorter blocking async save on 8×H200 (#7287).
- **DeepSpeed**: fixed a ZeRO-3 rollout deadlock and unsafe Hybrid Engine policy injection (#8264, #8265).

Compilers and kernels

- **PyTorch**: fixed an int32 index overflow in the `cdist` CUDA backward kernel (#198452).
- **FlashAttention**: block-sparse tiling and causal-mask skipping in the SM90 forward kernel (#2903, #2904).
- **Triton**: a nested-loop fusion miscompile (#11521) and fp64 atomic exchange on AMD GPUs (#11904).
- **LLVM**: ValueTracking inference for $x \neq y$ (#225200) and an Attributor crash fix (#225290).

Also: colibri (MoE expert streaming), JAX (batched `jax.scipy.linalg.lu`), Apache TVM (ONNX Hardmax with symbolic shapes), LMCache (allocator free-list corruption), and verl (FSDP critic loading).

Honors and Awards

National Encouragement Scholarship | People's Scholarship | Shanghai Municipal Scholarship
Social Practice Scholarship

Skills

Python, C, CUDA, Triton, SQL; PyTorch, JAX, vLLM, SGLang, DeepSpeed, Megatron-LM, MLIR, LLVM; NVIDIA and AMD GPUs.